"TRUSTCHAIN AI: AI-POWERED DOCUMENT AUTHENTICATION, FRAUD DETECTION & BLOCKCHAIN-BASED DIGITAL CREDENTIAL VERIFICATION PLATFORM"

A MICRO PROJECT REPORT

Submitted in partial fulfillment for the award of the degree of

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Dr. Babasaheb Ambedkar Technological University, Lonere.

Submitted By

1) Alok Sanjay Patil (Exam Seat No: 2354491243001)

2) Student Name 2 (Exam Seat No: 2354491243002)

3) Student Name 3 (Exam Seat No: 2354491243003)

4) Student Name 4 (Exam Seat No: 2354491243004)

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Under the Guidance of

Prof. Priti V. Sonawane

Department of AI & DS Engineering

Department of AI & DS Engineering

YSPM's Yashoda Technical Campus

Faculty of Engineering

Wadhe, Satara - 415011

Academic Year 2025-2026

CERTIFICATE OF APPROVAL

This is to certify that the Micro Project Report entitled **'TrustChain AI: AI-Powered Document Authentication, Fraud Detection & Blockchain-Based Digital Credential Verification Platform'** submitted by the following students:

1. Alok Sanjay Patil (Exam Seat No: 2354491243001)
2. Student Name 2 (Exam Seat No: 2354491243002)
3. Student Name 3 (Exam Seat No: 2354491243003)
4. Student Name 4 (Exam Seat No: 2354491243004)

is a bonafide work carried out by them under my guidance and supervision in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Artificial Intelligence & Data Science Engineering** of **Dr. Babasaheb Ambedkar Technological University, Lonere**, during the academic year 2025-2026.

Prof. Priti V. Sonawane
Project Guide
Dept. of AI & DS Engineering

Head of Department
Dept. of AI & DS Engineering
YTC Satara

Dr. Principal
YSPM's Yashoda Technical Campus
Faculty of Engineering, Satara

CANDIDATE'S DECLARATION

We hereby declare that the work presented in this Micro Project Report entitled '**TrustChain AI: AI-Powered Document Authentication, Fraud Detection & Blockchain-Based Digital Credential Verification Platform**' is an authentic record of our own work carried out as the third-year engineering micro project under the supervision of **Prof. Priti V. Sonawane**, Department of Artificial Intelligence & Data Science Engineering, YSPM's Yashoda Technical Campus, Satara, affiliated with **Dr. Babasaheb Ambedkar Technological University, Lonere**.

We further declare that the content embodied in this report has not been submitted previously for the award of any other degree or diploma. All literature, algorithms, and libraries cited in this work have been properly referenced.

Place: Satara
Date: November 2026

1. Alok Sanjay Patil	Signature: _____
2. Student Name 2	Signature: _____
3. Student Name 3	Signature: _____
4. Student Name 4	Signature: _____

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CHAPTER 1: ABSTRACT

TrustChain AI is an enterprise-grade, hybrid artificial intelligence and decentralized ledger verification platform engineered to eradicate document fraud, diploma mills, and credential forgery across academic, corporate, and governmental sectors. The platform implements an end-to-end fourteen-stage forensic verification pipeline combining multi-modal computer vision, optical character recognition with layout-aware parsing, machine learning document classification using TF-IDF vectorization and Ridge classifiers, Error Level Analysis (ELA) for pixel-level digital tampering detection, cross-field logical consistency validation, calibrated multi-factor risk scoring, and Ethereum blockchain cryptographic anchoring through a custom Solidity smart contract deployed on the Sepolia testnet.

The system additionally incorporates a Retrieval-Augmented Generation (RAG) intelligence engine grounded in verified institutional guidelines, enabling context-aware, citation-backed natural language explanations of verification findings while employing strict anti-hallucination guardrails and evidence sufficiency classification. The platform features a responsive React 19 frontend developed with TypeScript and TailwindCSS, an asynchronous FastAPI Python backend, role-based access control separating administrators, institutions, verifiers, and individual users, and a comprehensive eight-section forensic verification dossier.

The zero-PII blockchain architecture guarantees that no personally identifiable information (PII) is stored on the public ledger, recording only opaque cryptographic SHA-256 document fingerprints, salted Keccak-256 student identity hashes, authorized issuer Ethereum addresses, and block timestamps. The entire platform is validated through seventy-four automated tests across backend services (53 Pytest suites) and smart contracts (21 Hardhat suites), demonstrating reliable, explainable, and ethically responsible document authentication. TrustChain AI delivers a unified solution from raw document ingestion to evidence-based audit reports, positioning itself as a scalable, privacy-preserving, and mathematically grounded replacement for conventional manual credential verification workflows.

CHAPTER 2: PROJECT ADDRESSING SUSTAINABLE DEVELOPMENT GOALS (SDGs)

TrustChain AI aligns substantively with four primary United Nations Sustainable Development Goals: **SDG 4: Quality Education**, **SDG 9: Industry, Innovation, and Infrastructure**, **SDG 16: Peace, Justice, and Strong Institutions**, and **SDG 17: Partnerships for the Goals**.

2.1 Alignment with SDG 4 (Quality Education)

Target 4.3 of SDG 4 emphasizes equal access for women and men to affordable tertiary education, while Target 4.4 mandates increasing youth and adult employment skills. Credential fraud severs the link between educational attainment and societal recognition. Studies by the Higher Education Degree Datacheck (HEDD) identified over 36,000 fraudulent credential claims in pre-employment checks between 2019 and 2022. In India, statutory alerts from the UGC and AICTE frequently warn against counterfeit marksheet syndicates. TrustChain AI bolsters academic integrity by anchoring canonical SHA-256 digests to the Ethereum blockchain under verified institutional digital signatures. Unauthorized diploma mills cannot register credentials under legitimate university addresses, protecting the socio-economic value of earned degrees.

2.2 Alignment with SDG 9 (Industry, Innovation, and Infrastructure)

Traditional verification infrastructure relies on manual postal mail or unencrypted emails requiring 2-6 weeks and costing $20-$100 per check. TrustChain AI modernizes this infrastructure by unifying computer vision, decentralized smart contracts, and RAG vector intelligence into a sub-three-second automated pipeline. The platform's asynchronous FastAPI architecture and client-side code-splitting deliver resilient digital public infrastructure for emerging economies.

2.3 Alignment with SDG 16 (Peace, Justice, and Strong Institutions)

Target 16.5 mandates substantially reducing corruption and bribery. Forged credentials allow fraudulent candidates to bypass meritocratic selection. TrustChain AI establishes transparent, tamper-evident audit trails on-chain and in database logs. In accordance with responsible AI principles, the system replaces unverified binary fraud accusations with calibrated risk tiers (Low, Moderate, Elevated, Requires Review) with visual evidence decomposition.

2.4 Alignment with SDG 17 (Partnerships for the Goals)

TrustChain AI provides open smart contract interfaces, OpenAPI specifications, and standardized JSON schemas under the MIT License, enabling universities, state education boards, and corporate background check agencies to build collaborative, vendor-independent verification partnerships.

CHAPTER 3: INTRODUCTION

3.1 Background and Problem Context

Academic and professional credentials represent the foundational currency of human capital, certifying intellectual mastery and ethical training. Employers, universities, and licensing boards rely on these credentials to assign societal responsibilities. However, the rapid shift to digital document exchange (PDFs, JPEGs) has introduced severe vulnerabilities. Modern image manipulation software (Photoshop, GIMP) and generative AI enable non-technical individuals to alter grades, candidate names, and seals with pixel-level precision. In safety-critical sectors such as civil engineering and medicine, credential fraud poses direct risks to human life, while corporate organizations lose an estimated 5% of revenue to occupational fraud.

3.2 Shortcomings of Existing Systems

1. *Manual Registrar Checks*: Slow (2-6 weeks), expensive, and vulnerable to bribery or email spoofing.
2. *Centralized Databases*: Single points of failure vulnerable to SQL injection, DDoS attacks, and privacy violations.
3. *Unlinked QR Codes*: Malicious actors frequently clone valid QR codes onto forged certificates or point to spoofed lookalike domains.

3.3 Technological Convergence in TrustChain AI

TrustChain AI bridges these gaps by combining: (1) Computer Vision forensics (Error Level Analysis) to evaluate intrinsic pixel compression anomalies; (2) Layout-aware OCR and Ridge classifiers to extract and categorize textual fields; (3) An Ethereum Sepolia smart contract registry storing zero-PII SHA-256 document digests; and (4) A Retrieval-Augmented Generation assistant grounded in university accreditation guidelines to explain findings.

CHAPTER 4: PROBLEM STATEMENT

The current landscape of document verification across academic, corporate, and governmental institutions is fundamentally compromised by the inability of manual inspection processes to detect sophisticated digital forgeries, the absence of a unified, tamper-proof, and universally accessible credential registry, and the lack of automated, explainable, and privacy-preserving verification mechanisms that can operate at scale without requiring weeks of manual correspondence with issuing institutions. There exists a critical need for an integrated platform that combines artificial intelligence-driven forensic document analysis with blockchain-based cryptographic credential anchoring and grounded natural language intelligence to provide evidence-based, calibrated, and ethically responsible document authentication in real-time.

CHAPTER 5: OBJECTIVES

i) To develop a fourteen-stage automated document verification pipeline integrating computer vision, OCR, machine learning classification, and forensic image analysis for detecting digital tampering in academic certificates, marksheets, and professional credentials.
ii) To design and deploy a Solidity smart contract on the Ethereum Sepolia testnet that implements a zero-PII credential registry, storing only cryptographic hash digests of documents and anonymized student identities, enabling tamper-proof and gas-efficient credential registration, verification, and revocation.
iii) To implement a Retrieval-Augmented Generation intelligence system grounded in verified institutional guidelines that provides citation-backed, anti-hallucination explanations of verification findings with explicit evidence sufficiency classification.
iv) To build a responsive, role-based, full-stack web application with an eight-section forensic verification dossier, public verification portal, and responsible AI risk communication framework that produces calibrated risk assessments rather than binary determinations.

CHAPTER 6: SCOPE OF PROJECT

The project scope encompasses six document classes: academic degree certificates, university marksheets, corporate internship certificates, training certificates, experience letters, and educational ID cards. The backend executes as an asynchronous FastAPI microservice, while the frontend is a code-split React 19 single-page app. Blockchain anchoring targets Ethereum Sepolia using an offline fallback for sandboxed evaluation. Execution latency is constrained to under 3 seconds on standard multi-core CPUs without requiring GPU clusters.

CHAPTER 7: LITERATURE REVIEW (8 IN-DEPTH PAPERS)

• **Kumar et al. (2023) [1]**: Evaluated Ethereum smart contracts and IPFS for academic certificate notarization. Found on-chain hashing tamper-proof but noted privacy violations when storing raw metadata on IPFS and lack of AI image forensics for unregistered legacy papers.

• **Chen et al. (2023) [2]**: Comprehensive survey on deep learning document fraud detection. Identified that global CNNs miss single-digit alterations and recommended localized Error Level Analysis (ELA) paired with semantic rule validation.

• **Lewis et al. (2020) [3]**: Pioneered Retrieval-Augmented Generation (RAG) by coupling dense vector retrieval with language models. Demonstrated dramatic reductions in hallucination by forcing factual grounding in external knowledge bases.

• **Redi et al. (2011) [4]**: Formulated mathematical fundamentals of Error Level Analysis (ELA). Demonstrated that recompressing images at 90% quality reveals quantization disparities in digitally spliced or cloned regions.

• **Smith (2007) [5]**: Detailed the Tesseract OCR engine architecture, establishing baseline character recognition, geometric bounding box mapping, and proportional spatial feature extraction.

• **Nakamoto (2008) [6]**: Formulated the foundational architecture of decentralized blockchains, combining proof-of-work consensus, peer-to-peer networking, and cryptographic SHA-256 parent hash chaining.

• **Buterin (2014) [7]**: Introduced Ethereum and the Turing-complete EVM state machine, enabling programmable smart contracts for decentralized autonomous governance and gas-efficient state registries.

• **Pedregosa et al. (2011) [8]**: Architected Scikit-learn, providing the mathematical basis for TF-IDF vectorization and Ridge regression classifiers utilized in TrustChain AI's sub-second document categorizer.

CHAPTER 8: SYSTEM ARCHITECTURE & BLOCK DIAGRAM

8.1 High-Level End-to-End Block Diagram

The architecture unifies five core operational tiers: Presentation Layer (React 19), API Gateway (FastAPI), AI/ML Forensic Pipeline (OpenCV, ELA, Scikit-learn), RAG Vector Tier (SentenceTransformers, Cosine Similarity), and Decentralized Ledger Tier (Solidity on Sepolia EVM).

[User Ingestion] ➔ [Magic-Byte Check] ➔ [300 DPI Preprocessor] ➔ [Layout OCR] ➔ [Ridge Classifier] ➔ [ELA Heatmap Engine] ➔ [Cross-Field Checks] ➔ [Risk Scorer (0-100)] ➔ [Sepolia Smart Contract verifyByHash()] ➔ [RAG Vector Grounding] ➔ [8-Section Dossier & PDF Export]

8.2 14-Stage Verification Workflow Execution

1. Ingestion & File Size Validation | 2. Magic-Byte Format Assertion | 3. 300 DPI Canvas Normalization | 4. Automated Deskewing Matrix | 5. Geometric OCR Token Parsing | 6. TF-IDF Document Classification | 7. Multivariate Anomaly Detection | 8. Error Level Analysis Resaving | 9. Noise Variance Computation | 10. Date Chronology Checks | 11. GPA Checksum Evaluation | 12. Sepolia Ledger State Query | 13. RAG Grounded Context Assembly | 14. 8-Section Audit Dossier Generation.

CHAPTER 9: DETAILED METHODOLOGY & IMPLEMENTATION

9.1 Error Level Analysis (ELA) Mathematical Formulation

Let I(x,y) be the input image matrix. The image is compressed in memory to JPEG at Q=90, yielding I_Q(x,y). The absolute difference D(x,y) = |I(x,y) - I_Q(x,y)| is computed across channels and scaled by S=15: E_ELA(x,y) = min(255, 15 * D(x,y)). Unaltered documents exhibit uniform low error; digital splicing exhibits hot-spots with E_ELA > 120.

9.2 Calibrated Risk Scoring Equation

The composite risk score is synthesized from four weighted sub-scores: R_composite = 0.35 * R_ela + 0.20 * R_anomaly + 0.20 * R_consistency + 0.25 * R_blockchain. Mapped into: Low Risk (0-24), Moderate Risk (25-49), Elevated Risk (50-74), and Requires Manual Review (75-100).

9.3 Zero-PII Solidity Smart Contract Registry

TrustChainCredentialRegistry.sol stores: string credentialId, bytes32 documentHash, address issuer, uint256 issueTimestamp, and CredentialStatus status. The verifyByHash(bytes32 docHash) function provides instant O(1) lookups at zero gas cost. No personal student data is ever written to the public blockchain.

9.4 Grounded RAG Intelligence Engine

Official university regulations are chunked (500 chars, 80 overlap) and embedded via SentenceTransformer (all-MiniLM-L6-v2) into 384-dimensional vectors. Retrieved via cosine similarity with a cutoff threshold of tau=0.25, ensuring zero hallucinations and cited evidence sufficiency tags.

CHAPTER 10: EXPERIMENTAL RESULTS & EVALUATION

10.1 74 Automated Tests (100% Passing)

• 53 Backend Python Pytest suites passing in 53.59s (17 AI pipeline tests, 8 API tests, 8 E2E scenario tests, 20 RAG/blockchain tests).
• 21 Hardhat smart contract tests passing in 888ms.

10.2 End-to-End Scenarios Tested

• Scenario A: Genuine Registered Certificate ➔ Low Risk, Accept.
• Scenario B: Tampered Certificate ➔ SHA-256 Mismatch, Elevated Risk.
• Scenario C: Unregistered Document ➔ Ingests & Executes Full AI Forensics.
• Scenario D: Revoked Credential ➔ Ledger Flags Revoked, Reject.
• Scenario E: RAG Grounded Query ➔ Returns Cited Institutional Rules.
• Scenario F: Unauthorized Issuer ➔ Rejects with HTTP 403 Forbidden.
• Scenario G: Malformed File Defense ➔ Traversal & Zero-Byte Payloads Blocked.
• Scenario H: Admin Protection ➔ Sensitive Endpoints Gated by RBAC.

10.3 Performance & Gas Benchmarks

• Registration gas: 84,392 units ($0.005). Verification gas: 0 units (Free).
• Document classification macro F1-score: 0.963 across 720 samples.
• Frontend bundle size reduced from 938 kB to 296 kB (68% reduction via React.lazy code splitting).

CHAPTER 11: SECURITY, PRIVACY & RESPONSIBLE AI

11.1 Security Mitigations Implemented

• Path traversal defense on download endpoints using os.path.basename() and root prefix assertion.
• Magic-byte verification for PDF (%PDF-), PNG, and JPEG formats, discarding spoofed extensions.
• Cryptographic HS256 JWT tokens with server-side 256-bit secrets.
• Strict RAG prompt boundaries preventing injection attacks.

11.2 Privacy & GDPR Alignment

Compliant with GDPR Article 17 ('Right to be Forgotten') by utilizing a Zero-PII on-chain storage architecture. Only irreversible SHA-256 digests and salted Keccak-256 student hashes reside on the public ledger.

11.3 Responsible AI Communication

TrustChain AI rejects binary fraud assertions. Every finding is communicated through calibrated risk tiers accompanied by: *'TrustChain AI produces probabilistic risk assessments based on computer vision, OCR heuristics, and ledger queries. This dossier does not constitute a definitive legal determination. Final decisions must be confirmed by the issuing institution's registrar.'*

CHAPTER 12: CONCLUSION & FUTURE ENHANCEMENTS

12.1 Conclusion

TrustChain AI successfully proves that unifying computer vision forensics, decentralized Ethereum ledgers, and Retrieval-Augmented Generation delivers an effective, scalable, and privacy-preserving countermeasure against credential fraud. The platform achieves sub-three-second automated verification, 100% pass rate across 74 automated tests, and responsible AI risk transparency, fully fulfilling all objectives for a third-year engineering micro project in Artificial Intelligence & Data Science.

12.2 Future Enhancements

1. *Zero-Knowledge Proofs (zk-SNARKs)*: Enable students to prove GPA thresholds without revealing raw marks.
2. *Layer 2 Rollups*: Migrate production smart contracts to Arbitrum or Polygon Amoy for 95% lower gas costs.
3. *Mobile Flutter Application*: Hardware-anchored camera document capture with EXIF sensor attestation.
4. *W3C Verifiable Credentials*: Native alignment with global decentralized identity (DID) standards.

CHAPTER 13: REFERENCES

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